Problem Solving Using Python (CSE1012)

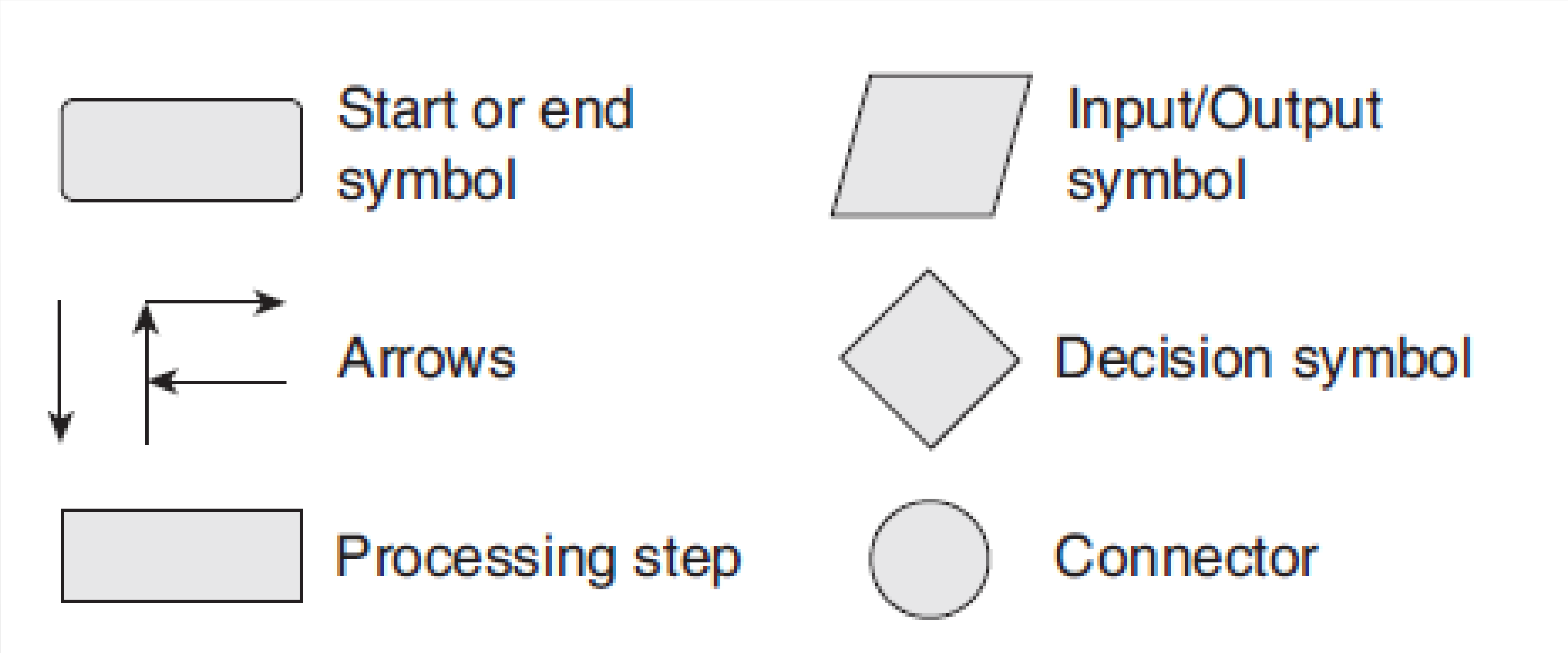
Dr. Abdul Kayom, Assistant Professor, SENSE

Agenda / Contents

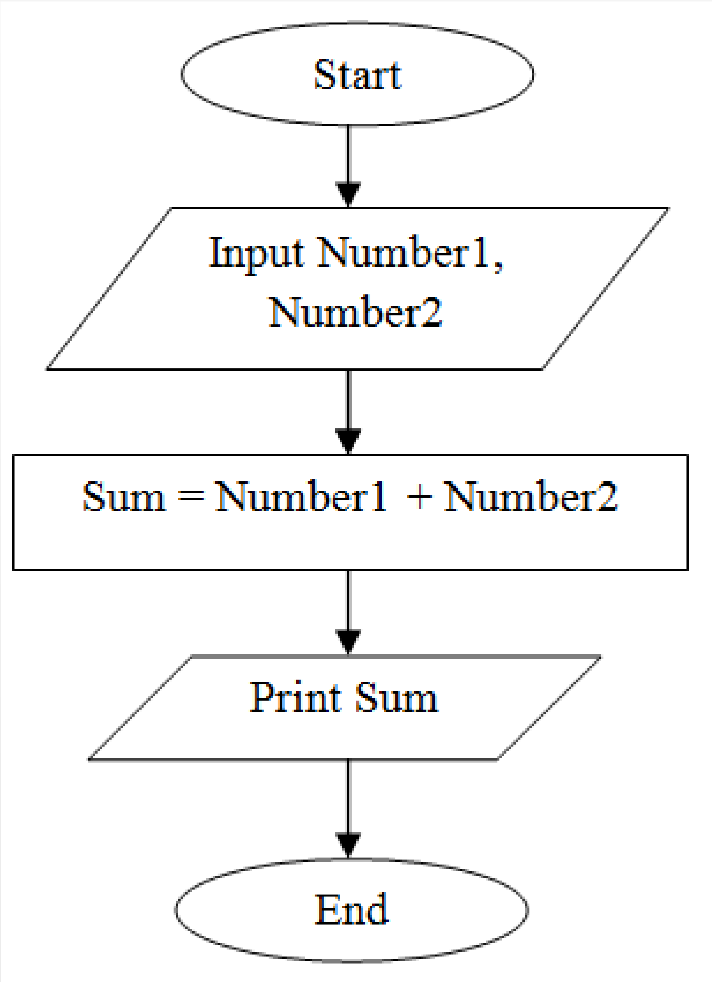
Flowchart

- **A flowchart is a graphical or symbolic representation of a process.**
- **It is basically used to design and document virtually complex processes to help the viewers to visualize the logic of the process.**

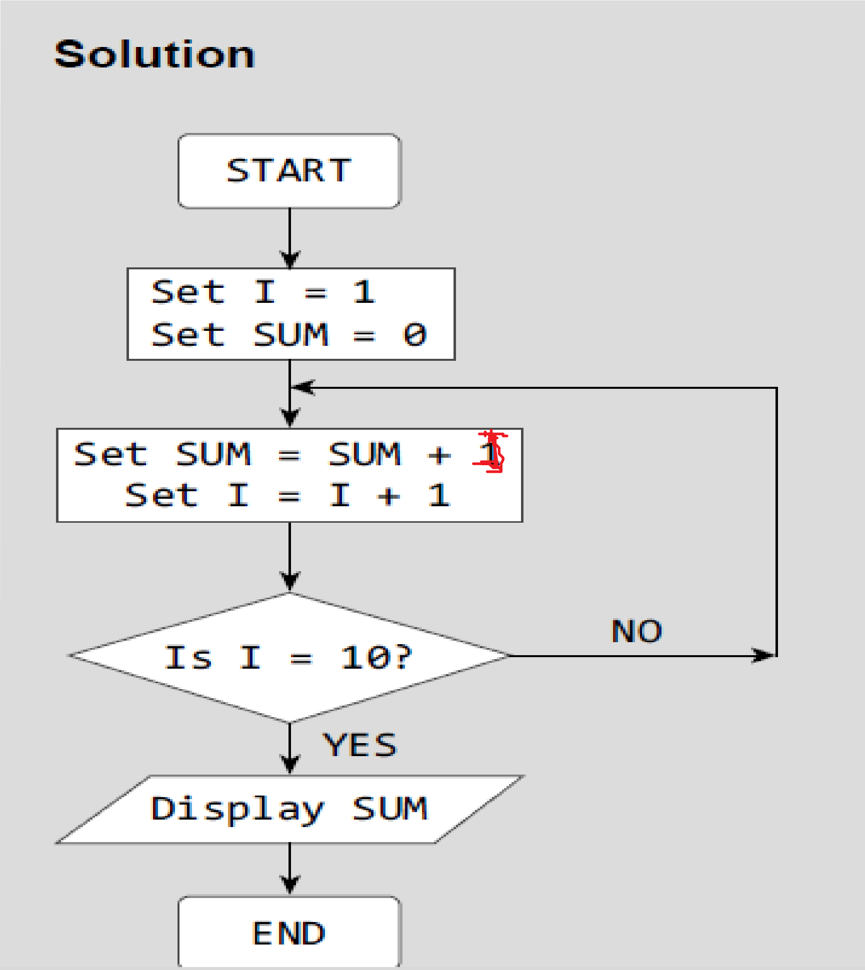
Flowchart


Start or end symbol
Input/Output symbol
Arrows
Decision symbol
Processing step
Connector

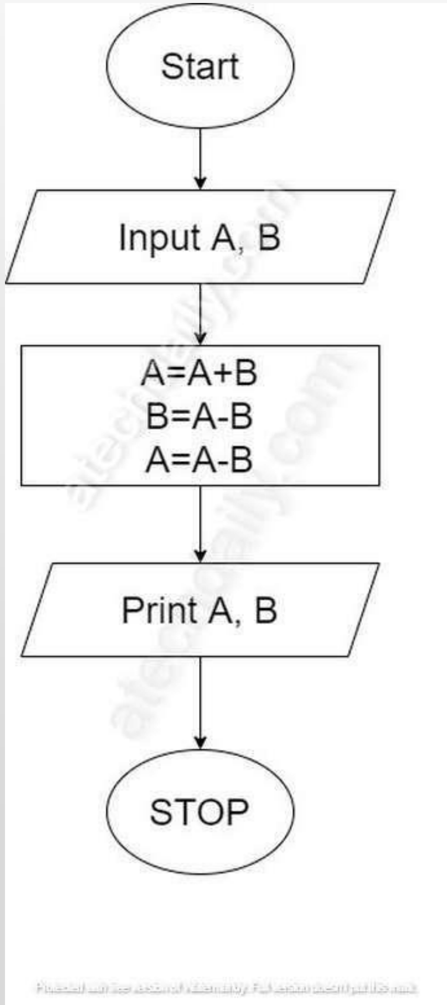
Flowchart: adding two numbers



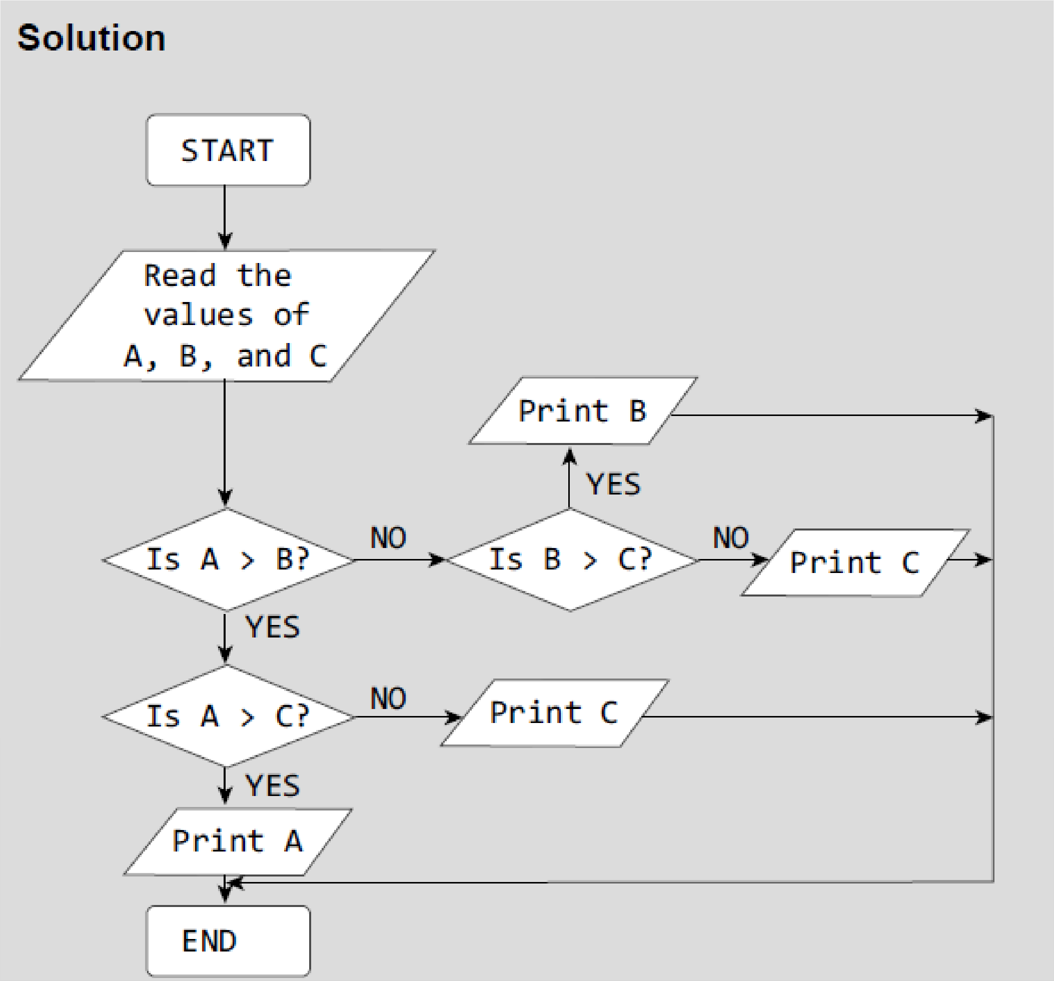
Flowchart: adding 10 natural numbers



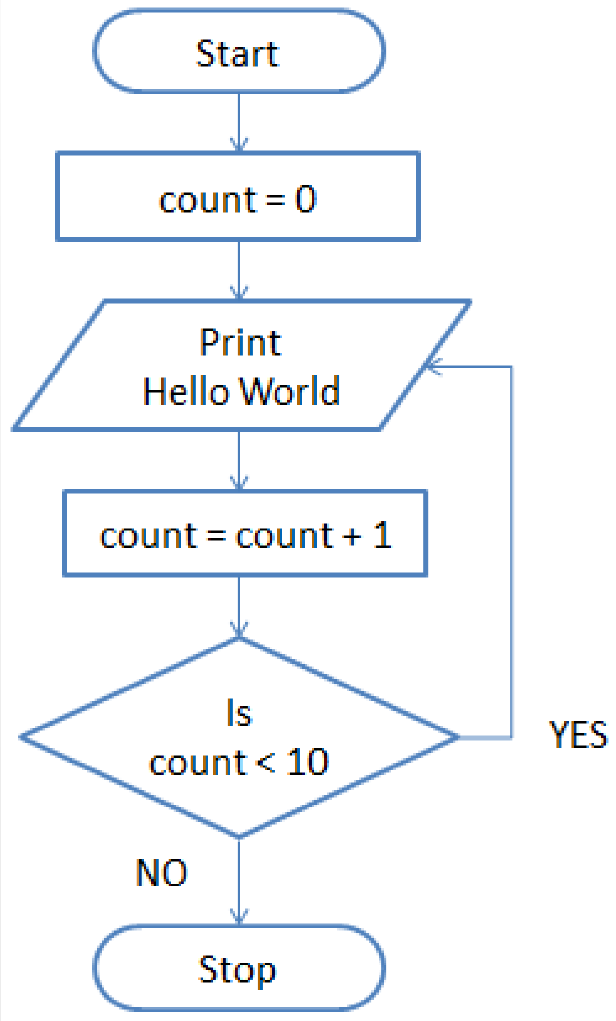
Interchange two values



Largest of Three Numbers



Count up to 10



Pseudocode

- **Pseudocode is a compact and informal high-level description of an algorithm that uses the structural conventions of a programming language.**

Pseudocode

- **It facilitates designers to focus on the logic of the algorithm without getting bogged down by the details of syntax.**
- **An ideal pseudocode must be complete, describing the entire logic of the algorithm, so that it can be translated straightaway into a program.**

Pseudocode

- Example: Write a pseudocode for calculating the price of a product after adding the sales tax to its original price.

Solution

```
1.  Read the price of the product
2.  Read the sales tax rate
3.  Calculate sales tax = price of the item x* sales tax rate
4.  Calculate total price = price of the product + sales tax
5.  Print total price
6.  End
Variables: price of the item, sales tax rate, sales tax, total price
```

Types of Errors

- **Run-time errors** occur when the program is being run/ executed.
- Such errors occur when the program performs some illegal operations like dividing a number by zero, opening a file that already exists, lack of free memory space, finding square or logarithm of negative numbers.
- Run-time errors may terminate program execution.

Types of Errors

- **Syntax Errors** are generated when rules of a programming language are violated.
- Python interprets (executes) each instruction in the program line by line.
- The moment interpreter encounters a syntactic error, it stops further execution of the program.

Types of Errors

- **Semantic or Logical Errors** are those errors which may comply with rules of the programming language but gives an unexpected and undesirable output which is obviously not correct.
- For example, if you write a program to add two numbers but instead of writing ‘+’ symbol, you put the ‘-’ symbol.

Types of Errors

- **Linker Errors These errors occur when the linker is not able to find the function definition for a given prototype.**

Testing

- **Unit Tests:** **Unit testing is applied only on a single unit or module to ensure whether it exhibits the expected behavior.**

Testing

- **Integration Tests:** **These tests are a logical extension of unit tests. In this test, two units that have already been tested are combined into a component and the interface between them is tested.**

Testing

- **System Tests:** **System testing checks the entire system.**

Debugging

- ***Debugging*** **is an activity that includes execution testing and code correction.**
- **It locates errors in the program code and fix them to produce an error-free code. Different approaches applied for debugging a code includes:**

Debugging

- **Brute-Force Method:** **A printout of CPU registers and relevant memory locations is taken, studied, and documented.**
- **It is the least efficient way of debugging a program and is generally done when all the other methods fail.**

Debugging

- **Backtracking Method:** It is used to debug small applications.
- It works by locating the first symptom of error and then tracing backward across the entire source code until the real cause of error is detected.

Debugging

- **Backtracking Method:** **With increase in number of source code lines, the possible backward paths become too large to manage.**

Debugging

- **Cause Elimination:** **A list of all possible causes of an error is developed.**
- **Then relevant tests are carried out to eliminate each of them.**

Thank You

Dr. Abdul Kayom

Asst. Prof, (SENSE)

kayomabdul@vitap.ac.in

8474860025